

Consumer Shopping Behavior Analysis

Business Report

Executive Summary

Consumer behavior analysis is essential for retail organizations seeking to improve customer satisfaction, increase sales, and enhance long-term customer loyalty. This project focuses on analyzing consumer shopping data to understand purchasing patterns, customer preferences, and factors influencing buying decisions. The analysis was conducted using Python for data preprocessing, PostgreSQL for database management, and Power BI for interactive visualization.

The dataset contains information related to customer demographics, product categories, purchase amounts, discounts, customer ratings, payment methods, sales channels, and loyalty status. Through data cleaning, transformation, and analysis, valuable insights were extracted regarding customer spending behavior, product performance, and sales trends.

The study revealed that customer demographics, discounts, product ratings, and sales channels significantly influence purchasing decisions. Loyalty program members demonstrated stronger repeat purchase behavior, while specific product categories generated higher revenue compared to others. Additionally, online sales channels showed increasing customer engagement, highlighting the importance of digital retail strategies.

Based on these findings, several business recommendations were proposed, including strengthening loyalty programs, implementing personalized marketing campaigns, optimizing discount strategies, and focusing on high-performing product categories. These recommendations can help the organization improve customer retention, increase profitability, and support data-driven decision-making.

Overall, this project demonstrates how data analytics can transform raw consumer shopping data into actionable business insights, enabling organizations to enhance operational efficiency and gain a competitive advantage in the retail industry.